INSTITUT FOR MATEMATIK
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October 22, 2026

Statistical Models
Week 9: Objective priors and numeric methods

comments

> **!** Extended lectures on Wednesday
>
> Wednesday this week (October 29th), and Wednesday in week 11 (November 12th), we will have longer of lectures, that is from 14-17.
> This is to make up for the fact that there will be no lectures on Wednesday, November 5th (course week 10).

Topics for this week include:

- ☐ More on priors: examples for conjugate priors, mixtures of conjugate priors, and objective priors, in particular Jeffreys' prior.
- ☐ Calculating summaries for the posterior when the prior is not the conjugate prior: calculation of maximum a posteriori estimate (MAP), numerical integration, and Laplace's approximation for the posterior: the "Bayesian CLT".
- ☐ If time allows, we have a look at the paper by Mr.Bayes and Mr.Price (1763): "An Essay towards Solving a Problem in the Doctrine of Chances", *Philosophical Transactions*, 1763, Vol. 53, pp. 370-418, where they introduced the idea of sequentially updating one's believe by experiments.
- ☐ Finally, we will discuss the general idea of Markov Chain Monte Carlo methods. These methods are based on results for Markov chains, that we will review. You may know Markov chains with finitely many possible outcomes, that is, the values are integers (finite state space). Here, we use Markov chains with real valued outcomes (continuous state space).

Relevant Readings

- More on priors: R&G, Sections 2.1, 2.3.1
- Numerical methods: R&G, Section 3.1
- Notes on Markov chains with finite and continuous state space (may get updated before we use them)

Lecture Material

- Link to onenotes on Google drive
- Code for R&g, Figure 2.5
- Code for R&G, Figure 3.2
- Experiments with AR(1) chains

Exercises

tba